

APPLETON, WI, USA

WMO: 726457

Lat: 44.267N

Lon: 88.517W

Elev: 280

StdP: 98.01

Time Zone: -6.00 (NAC)

Period: 92-19

WBAN: 04825

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.1	-17.9	-25.8	0.4	-20.4	-22.6	0.5	-17.1	12.4	-8.0	11.1	-5.2	4.7	290	0.604

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.8	31.3	23.9	29.1	22.4	27.9	21.4	25.3	29.5	24.1	28.0	23.0	26.5	5.3	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.9	19.5	27.5	22.8	18.1	26.3	22.1	17.4	25.6	78.7	29.8	73.9	27.8	69.8	26.1	33.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.1	9.7	8.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.3	-7.2	-4.1	0.4	8.0	13.5	19.3	22.2	20.8	17.1	9.8	2.8	-3.9
	DBStd	11.22	6.00	6.22	5.78	4.90	4.60	4.19	3.07	3.17	4.27	4.93	5.20	6.51
	HDD10.0	2061	533	394	302	93	18	0	0	0	3	64	222	433
	HDD18.3	4006	791	627	556	312	162	40	6	13	74	267	467	690
	CDD10.0	1432	0	0	4	33	126	278	377	334	215	59	5	1
	CDD18.3	334	0	0	0	2	12	68	124	89	37	4	0	0
	CDH23.3	2504	0	0	1	24	111	544	987	571	239	27	0	0
	CDH26.7	669	0	0	0	4	23	152	296	146	44	3	0	0
Wind	WSAvg	4.3	4.8	4.6	4.4	5.0	4.6	3.9	3.6	3.3	4.0	4.3	4.5	4.5
Precipitation	PrecAvg	810	34	26	46	72	95	119	99	90	78	63	46	39
	PrecMax	1049	70	45	93	135	192	227	291	234	149	147	107	83
	PrecMin	708	7	7	5	16	41	37	28	42	15	7	5	5
	PrecStd	92	18	11	26	28	38	56	63	41	34	36	29	23
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.7	11.4	19.1	26.4	29.1	32.6	34.1	32.5	30.0	27.1	18.7	13.8
		MCWB	4.6	10.4	13.1	16.2	20.0	23.0	26.8	25.4	22.5	20.5	13.6	12.3
	2%	DB	4.1	7.5	14.2	22.5	26.9	30.2	32.0	30.0	27.7	23.7	15.9	9.8
		MCWB	1.8	5.0	10.0	14.8	19.0	22.7	24.7	23.6	20.9	18.0	12.0	9.2
	5%	DB	2.3	5.5	12.1	18.8	23.8	28.2	29.9	27.8	26.1	21.1	12.8	6.3
		MCWB	1.4	3.4	8.8	13.0	17.0	21.8	23.4	22.0	19.6	16.9	9.3	4.3
	10%	DB	1.1	3.7	8.4	16.3	21.9	27.1	28.0	27.0	23.9	17.6	11.0	3.8
		MCWB	0.5	2.2	6.0	9.9	15.3	20.8	21.5	21.2	18.9	14.0	8.5	2.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.9	10.3	14.8	18.8	22.3	25.7	28.8	26.6	23.7	21.5	14.6	12.5
		MCDB	7.9	11.9	17.3	24.1	27.1	30.1	33.0	30.5	27.4	25.1	17.5	13.6
	2%	WB	2.5	5.4	11.9	16.0	20.0	24.1	25.8	24.6	22.5	19.4	12.7	8.6
		MCDB	4.0	7.2	13.2	20.9	24.8	28.6	29.6	28.5	26.0	22.8	15.0	9.4
	5%	WB	1.4	3.6	8.6	13.7	18.3	22.8	24.5	23.5	21.2	17.1	10.4	4.6
		MCDB	2.3	5.2	11.1	17.8	22.5	27.4	28.4	26.8	24.5	19.7	12.4	5.9
	10%	WB	0.2	2.4	6.0	11.2	16.9	21.6	23.3	22.2	20.0	14.7	8.0	2.6
		MCDB	1.0	3.7	8.8	15.5	20.2	25.6	26.6	25.3	23.4	18.0	10.0	4.0
Mean Daily Temperature Range	5% DB	MDBR	7.4	7.5	8.6	10.5	9.9	10.0	9.8	9.2	10.3	9.4	7.7	7.2
		MCDBR	8.0	8.4	12.5	15.1	12.7	11.4	11.2	10.5	11.5	13.0	11.3	8.8
		MCWBR	6.9	6.6	8.5	9.0	7.0	5.9	5.6	5.7	5.8	7.7	8.2	7.2
	5% WB	MCDBR	7.4	8.1	12.3	13.3	10.9	10.5	10.1	9.4	10.1	10.3	10.1	8.1
		MCWBR	6.6	6.5	9.4	9.3	7.2	6.4	5.7	6.1	6.0	7.6	8.6	7.2
Clear-Sky Solar Irradiance	taub	0.265	0.320	0.330	0.370	0.400	0.406	0.392	0.393	0.365	0.325	0.294	0.270	
	taud	2.402	2.217	2.363	2.311	2.269	2.288	2.331	2.322	2.391	2.503	2.516	2.444	
	Ebn at Noon	871	857	906	893	871	865	872	859	857	855	832	828	
	Edh at Noon	73	107	106	122	131	130	123	120	103	81	67	65	
All-Sky Solar Radiation	RadAvg	1.63	2.70	3.85	4.55	5.47	5.98	6.13	5.23	4.17	2.64	1.69	1.22	
	RadStd	0.19	0.26	0.41	0.46	0.43	0.43	0.28	0.34	0.33	0.27	0.15	0.12	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page